

Lecture 10.

Theorem:- Mathematical statement which can be proved.

Axiom / Postulate:- A statement that is assumed to be true without a proof.

Corollary:- An easy consequence of a theorem.

Conjecture:- A statement believed to be true based on observation yet unproven.

Direct:-

$$P \rightarrow Q.$$

Definition:- Even:- $n \in \mathbb{Z}$ is even if $\exists k \in \mathbb{Z} \quad n = 2k$.
Odd:- $n \in \mathbb{Z}$ is odd if $\exists k \in \mathbb{Z} \quad n = 2k+1$.

Ex 1:-
PTS

if n is odd then n^2 is also odd.

Assume n is odd is true.

then According to def $n = 2k+1$. ^① For $k \in \mathbb{Z}$.

Taking Square both sides.

$$\begin{aligned} n^2 &= (2k+1)^2 = 4k^2 + 1 + 4k \\ &= 4k^2 + 4k + 1 \\ &= 2(2k^2 + 2k) + 1. \end{aligned}$$

$L = 2k^2 + 2k$ is also Integer.

$$L \in \mathbb{Z}.$$

$$= 2L+1.$$

which is also odd.

Implies n^2 is also odd.

Definition:- $a \in \mathbb{Z}$ is a perfect Square
if $\exists b \in \mathbb{Z}$ such that $a = b^2$.

Ex 2:- if m, n are perfect Squares then
 mn is also perfect Square.

Solution:-

If m & n are perfect Squares then
According to def.

$$\exists a \in \mathbb{Z} \quad m = a^2 \quad \text{--- (1)}$$

$$\exists b \in \mathbb{Z} \quad n = b^2 \quad \text{--- (2)}$$

King Rights sides & left sides of (1) & (2).

$$mn = a^2 b^2$$

$$mn = (ab)^2.$$

$$\text{if } a \in \mathbb{Z} \wedge b \in \mathbb{Z} \\ ab \in \mathbb{Z}.$$

Hence mn is also a perfect Square.

INDIRECT proofs,

1- Proof by Contraposition.

2- Proof by Contradiction. $P \rightarrow Q \equiv \neg Q \rightarrow \neg P.$

Ex 3 :- if $n \in \mathbb{Z}$ $3n+2$ is odd then n is odd.
PT6 P Q

Consider $\neg Q$ to be true.

n is even. then According to def.

$$n = 2K \quad \text{--- (1) } K \in \mathbb{Z}.$$

Substitute (1) in $3n+2$

$$n = 2k \text{ --- (1) } 1 \in \mathbb{Z}.$$

Substitute.

$$(1) \text{ in } 3n+2.$$

$$= 3(2k)+2.$$

$$= 6k+2.$$

$$= 2(3k+1).$$

Since $3k+1 \in \mathbb{Z}$.

$$= 2(\text{Integer}) = \text{Even}.$$

which means $3n+2$ is even.

which contradicts hypothesis.

Hence proved.

P.

Ex 4:-
P76.

$$(1) \text{ } (n = ab) \text{ } (a \in \mathbb{Z}^+, b \in \mathbb{Z}^+).$$

$$\text{then } (a \leq \sqrt{n} \vee b \leq \sqrt{n}).$$

$$\neg q = \neg(a \leq \sqrt{n} \vee b \leq \sqrt{n})$$

$$\Rightarrow a > \sqrt{n} \wedge b > \sqrt{n}. \text{ is true.}$$

$$ab > \sqrt{n} \cdot \sqrt{n}$$

$$ab > n.$$

(this is because $a > 0$ & $b > 0$)

$$\Rightarrow ab \neq n.$$

which is contradiction of hypothesis

Hence proved.

Proof by Contradiction.

$$p \rightarrow \neg q.$$

Def:- Rational:-

$x \in \mathbb{R}$ is rational $\exists p, q \in \mathbb{Z}, q \neq 0$ $x = p/q$.
 otherwise irrational.
 $p \nmid q$ have no common factors.

Example:- $\sqrt{2}$ it is irrational.
 Assume

$\sqrt{2}$ is rational.

There must exist $a, b \in \mathbb{Z}$ such that
 $\sqrt{2} = \frac{a}{b}$ — (1) $b \neq 0$.

Taking Square $2 = \frac{a^2}{b^2} \Rightarrow 2b^2 = a^2$ — (2).

Which means a is even. $\therefore a = 2k$
 $a = 2k$ — (A) $k \in \mathbb{Z}$. $\forall \exists k \in \mathbb{Z}$.

Put (A) in (2)

$$\begin{aligned} 2b^2 &= (2k)^2 \\ 2b^2 &= 4k^2 \\ b^2 &= 2k^2 \end{aligned}$$

b^2 is also even.

this means $\sqrt{2} = \frac{a}{b}$ is not in Simplest form

which means it's not because a & b have a common factor.

which means our supposition was

incorrect
 hence proved.

$\sqrt{2}$ is irrational.